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1: function BS23_ADAPTIVE( $f, [t_0, t_f], x_0, E_s$ )
2:    $t \leftarrow [t_0], \quad x \leftarrow [x_0], \quad h \leftarrow (t_f - t_0)/100$ 
3:    $k_1 \leftarrow f(t_0, x_0)$ 
4:   while  $t_{\text{end}} < t_f$  do
5:      $h \leftarrow \min(h, t_f - t_{\text{end}})$ 
6:      $k_2 \leftarrow f(t_{\text{end}} + \frac{h}{2}, x_{\text{end}} + \frac{h}{2}k_1)$ 
7:      $k_3 \leftarrow f(t_{\text{end}} + \frac{3h}{4}, x_{\text{end}} + \frac{3h}{4}k_2)$ 
8:      $x_3 \leftarrow x_{\text{end}} + h \left( \frac{2}{9}k_1 + \frac{1}{3}k_2 + \frac{4}{9}k_3 \right)$  ▷ 3rd order
9:      $k_4 \leftarrow f(t_{\text{end}} + h, x_3)$ 
10:     $x_2 \leftarrow x_{\text{end}} + h \left( \frac{7}{24}k_1 + \frac{1}{4}k_2 + \frac{1}{3}k_3 + \frac{1}{8}k_4 \right)$  ▷ 2nd order
11:     $E_a \leftarrow |x_3 - x_2|$ 
12:    if  $E_a \leq E_s$  then
13:      append  $t_{\text{end}} + h$  to  $t$ 
14:      append  $x_3$  to  $x$ 
15:       $k_1 \leftarrow k_4$  ▷ slope at the new  $x_{\text{end}}$ 
16:    end if
17:     $h \leftarrow 0.9 h (E_s/E_a)^{1/3}$  ▷ b/c  $E_a$  is  $O(h^3)$ 
18:  end while
19:  return  $t, x$ 
20: end function

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